11.4.2 Client certificate authentication

To enable TLS client certificate authentication for service clients you need to configure the server to send a *CertificateRequest* message as part of the handshake and to validate any certificate that it receives. Most application servers and reverse proxies support configuration options for requesting and validating client certificates, but these vary from product to product. In this section you'll configure the Nginx ingress controller from chapter 10 to allow client certificates and verify that they are signed by a trusted CA.

To enable client certificate authentication in the Kubernetes ingress controller, you can add annotations to the ingress resource definition in the Natter project. Table 11.1 shows the annotations that can be used.

> **NOTE** All annotation values must be contained in double quotes, even if they are not strings. For example, you must use `nginx.ingress.kubernetes.io/auth-tls-verify-depth: "1"` to specify a maximum chain length of 1.

Table 11.1 Kubernetes Nginx ingress controller annotations for client certificate authentication

Annotation	Allowed values	Description
`nginx.ingress.kubernetes.io/auth-tls-verify-client`	`"on"`, `"off"`, `"optional"`, or `"optional_no_ca"`	Enables or disables client certificate authentication. If "on" then a client certificate is required. The "optional" value requests a certificate and verifies it if the client presents